4.4

1(2)

$$y = \frac{1}{1-x}$$

$$y' = \frac{-\cdot(-1)}{(1-x)^2} = \frac{1}{(1-x)^2}$$

$$\therefore \quad y = 1\cdot(x-0)+1 = x+1$$

2(2)

$$f(x) = \frac{1}{x}$$

$$f(9.78) = \frac{1}{9.78}$$

9.78 근방의 정수 : 10 $x_0 = 10$ 에서의 $L(x)$ 구하면

$$f'(x) = -\frac{1}{x^2}$$

$$\therefore \quad L(x) = -\frac{1}{100}(x-10) + \frac{1}{10}$$

$$L(9.78) = -\frac{1}{100}\cdot(-0.22) + \frac{1}{10}$$

$$= 0.0022 + 0.1$$

$$= 0.1022 \text{ //}$$

5

$$f(x) = x^4 - 8x^2 + 75$$

$$f'(x) = 4x^3 - 16x$$

$$f''(x) = 12x^2 - 16$$

$f'(x) = 0$ 인 x는 $4x(x^2-4) = 0$ $0, \pm 2$ 이다.

$f''(0) = -16 < 0$ (극대)

$f''(2) = 32 > 0$ (극소)

$f''(-2) = 32 > 0$ (극소)

6(3)

$$y = \frac{x}{x^2+1}$$

$$dy = f'(x)\,dx$$

$$= \frac{x^2+1-2x^2}{(x^2+1)^2}dx = \frac{-x^2+1}{(x^2+1)^2}dx$$